(ENG) Kinematics: velocity

Speed and velocity

It is not the same to move fast (speed) as to know in which direction you are moving with that speed (velocity).

- **Speed**: how fast something moves. Unit: $\frac{m}{s}$.
- **Velocity**: speed + direction of that motion. It is also measured in $\frac{m}{s}$ and the sign will indicate whether it goes to the right (+) or to the left (-).

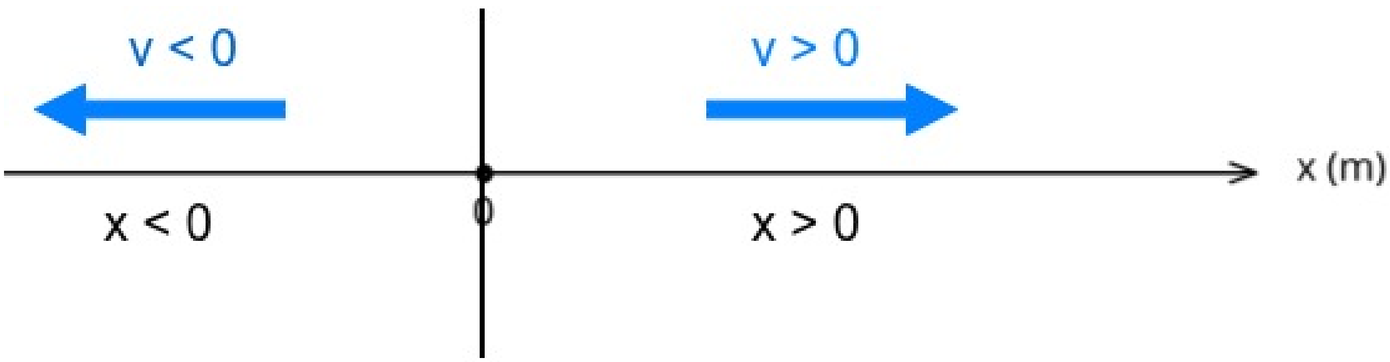

Figura 1: Sign convention for position and velocity

The *direction* indicates whether it goes to the right (positive velocity) or to the left (negative velocity) according to the given sign convention. It is represented with an arrow. The length of that arrow, which is the *magnitude* (the numerical value without sign), would be the speed.

Normally we will use the term **velocity**, but this implies that we must be careful with the signs we obtain:
- The displacement Δx can be positive or negative.
- **Time will always be positive**.

❗ Very important

Solving the formula for different variables

It is very important to know how to calculate velocity, and also how to solve the formula for all variables.

- Velocity:
  $v = \frac{s}{t}$. It is measured in **meters per second** ($\frac{m}{s}$)
- Solving for distance from velocity:
  $v = \frac{s}{t} \rightarrow v \cdot t = s$. It is measured in **meters (m)**.
- Solving for time from velocity:
  $v = \frac{s}{t} \rightarrow v \cdot t = s \rightarrow t = \frac{s}{v}$. It is measured in **seconds (s)**.

Units of velocity: $\frac{m}{s}$ and $\frac{km}{h}$

In general it is better to write $\dfrac{m}{s}$ instead of m/s and $\dfrac{km}{h}$ instead of km/h because this way we clearly know what the denominator is.

It is very useful to learn how to convert between different velocity units and we will do it using conversion factors:

Conversion from $\dfrac{m}{s}$ to $\dfrac{km}{h}$

$$1\frac{m}{s} = \frac{1\ m}{s} \cdot \frac{1\ km}{1000\ m} \cdot \frac{60 \cdot 60\ s}{1\ h} = \frac{3600\ km}{1000\ h} = 3{,}6\frac{km}{h}$$

Conversion from $\dfrac{km}{h}$ to $\dfrac{m}{s}$

$$1\frac{km}{h} = \frac{1\ km}{h} \cdot \frac{1000\ m}{1\ km} \cdot \frac{1\ h}{3600\ s} = \frac{1000\ m}{3600\ s} = \frac{1}{3{,}6}\frac{m}{s}$$

Unit conversion exercises

1. Convert $15m/s$ into km/h.
2. Convert $90km/h$ into m/s.

Solutions

1. Convert $15m/s$ into km/h.

$$15\frac{m}{s} \cdot \frac{1\,km}{1000\,m} \cdot \frac{3600\,s}{1\,h} =$$
$$= 15 \cdot \frac{3600}{1000}\frac{km}{h} =$$
$$= 15 \cdot 3.6 = 54\frac{km}{h}$$

- *Result:* $15\dfrac{m}{s} = 54\dfrac{km}{h}$.

2. Convert $90km/h$ into m/s.

$$90\frac{km}{h} \cdot \frac{1000\,m}{1\,km} \cdot \frac{1\,h}{3600\,s} =$$
$$= 90 \cdot \frac{1000}{3600}\frac{m}{s} =$$

$$= \frac{90}{3.6} = 25\frac{m}{s}$$

- *Result:* $90\frac{km}{h} = 25\frac{m}{s}$

Velocity problems

1. The Castellana Bus. An EMT bus travels the **6 km** between Plaza de Colón and the Cuatro Torres. If the journey is done in a straight line (without stops) and takes exactly **15 minutes** (0.25 hours), what is the average velocity of the bus in km/h?

Solution

We apply the formula: $v = \frac{d}{t}$

$v = \frac{6 \text{ km}}{0,25 \text{ h}} = 24 \text{ km/h}$

2. The scooter in Retiro Park. A boy travels along Paseo de Fernán Núñez in Retiro Park, which has a length of **750 meters**, on his electric scooter. If it takes him **150 seconds** to complete the route at constant speed, what is his velocity in meters per second (m/s)?

Solution

$v = \frac{d}{t}$

$v = \frac{750 \text{ m}}{150 \text{ s}} = 5 \text{ m/s}$

3. The Madrid Metro (Line 1). A metro train is at position $x_i = 200$ m (taking Puerta del Sol as the origin). A few seconds later, it is at position $x_f = 1400$ m (near Bilbao station). If it took **80 seconds** to move from one point to the other, first calculate the displacement and then its velocity in m/s.

Solution

1. We calculate the displacement (Δx): $x_f - x_i = 1400 - 200 = 1200$ m
2. We calculate the velocity: $v = \frac{\Delta x}{t} = \frac{1200 \text{ m}}{80 \text{ s}} = 15$ m/s

4. Shopping on Gran Vía. Two friends are walking. They start at number 1 on Gran Vía (position 0 m) and walk to Plaza de España, which is at position 1300 m. If they take **20 minutes** (1200 s) to arrive, calculate the displacement and the average velocity in m/s.

Solution

1. Displacement: $1300 \text{ m} - 0 \text{ m} = 1300 \text{ m}$

2. Velocity: $v = \dfrac{1300 \text{ m}}{1200 \text{ s}} \approx 1,08 \text{ m/s}$

5. The commuter train to Alcalá de Henares A commuter train travels from Atocha station to Alcalá de Henares with a constant velocity of **90 km/h**. If the journey lasts **20 minutes** (that is, $1/3$ of an hour or $0,33$ h), what distance has the train traveled?

Solution

1. We write the velocity equation: $v = \dfrac{d}{t}$.
2. We solve for displacement from the previous equation (since t is dividing, we move it multiplying): $v \cdot t = d \rightarrow d = v \cdot t$
3. We calculate with numerical values: $d = 90 \text{ km/h} \cdot 0,333 \text{ h} = 30 \text{ km}$

6. Trip to Segovia by car A family decides to go from Madrid to Segovia by highway. The distance is **90 km**. If they maintain a constant average velocity of **120 km/h** (the legal maximum), how long will it take them to reach their destination in hours and minutes?

Solution

1. We write the velocity equation: $v = \dfrac{d}{t}$.
2. To avoid mistakes, first we eliminate the denominator t and then solve for t. **Be very careful, otherwise you may get it reversed!!**
$$v = \frac{d}{t} \rightarrow v \cdot t = d \rightarrow t = \frac{d}{v}$$
3. We substitute numerical values and calculate: $t = \dfrac{90 \text{ km}}{120 \text{ km/h}} = 0,75$ hours
4. To convert to minutes, we take the non-integer part (the integer part is hours) and multiply by 60 because each hour has 60 minutes:
$$t = 0,75h \rightarrow 0,75h \cdot \frac{60min}{1h} = 45 \text{ minutes}$$

Recursos adicionales

- English video Motion in one dimension
- Movimiento rectilíneo